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Reflection loss mechanism of single layer absorber for flake-shaped carbonyl-iron particle composite

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The reflection loss (RL) properties including the frequency, intensity, and bandwidth of absorption peaks for the flake-shaped carbonyl-iron particle/paraffin composite were investigated in detail. By measuring the reflection coefficient of the composite without ($S_{11\text{-OPEN}}$) and with ($S_{11\text{-SHORT}}$) a backed metal plate, the origin of the RL peak was directly verified. The separated electromagnetic wave energy reflected at the air-absorber interface and the absorber-metal plate interface was achieved from $S_{11\text{-OPEN}}$ and $S_{11\text{-SHORT}}$ parameters, and they were used to explain the intensity variation of the RL peak. The bandwidth of the RL peak was also deduced from viewpoint of interface reflection waves. The calculated bandwidth from the derived formula agreed well with the experimental data. © *2012 American Institute of Physics*. [http://dx.doi.org/10.1063/1.4767365]

I. INTRODUCTION

Electromagnetic interference and radiation in GHz range become more serious with the development of computer and communication techniques. To solve these problems, the absorber can be used to delete the additional electromagnetic wave. The electromagnetic wave absorber filled by flake-shaped metallic magnetic particles with high saturation magnetization, such as Fe, FeNi, and FeSiAl particles, have attracted much attention during recent years besides the dielectric materials with light weight.[1–4] As the magnetic moments lie in the flake plane, the product of initial susceptibility and natural resonance frequency can exceed Snoek limit and hence the flake-shaped particles can possess high permeability in GHz range.[5] By using this kind of magnetic particles as fillers, a thinner absorber can be achieved.[6] In previous research on absorbers filled by flake-shaped metallic magnetic particles, most of their studies only paid attention to the dependence of RL peak frequency on absorber thickness, and few studies were put into explaining the intensity variation of RL peak and the bandwidth. As we know, the electromagnetic wave absorption performance strongly depends on the frequency, intensity, and bandwidth of RL peak. A complete understanding of these properties is vital to recognize the RL mechanism and design the absorber with high performance. In this work, the frequency, intensity, and bandwidth of RL peaks for flake-shaped carbonyl-iron (FSCI) particle/paraffin composites were deeply investigated from viewpoint of their origin.

II. EXPERIMENT AND CHARACTERIZATION

The FSCI particles shown in Fig. 1 were fabricated in our laboratory. The detailed fabrication method and experimental conditions have been reported elsewhere.[7] The magnetic hysteresis loop of the particles at room temperature, shown in Fig. 2, exhibits that the sample has excellent soft magnetic properties. The paraffin composites were prepared by mixing particles and paraffin with 20% volume concentration of particles. The composites for microwave measurement were pressed into a toroidal-shape with an outer diameter of 7.00 mm and an inner diameter of 3.04 mm at different thickness from 0.65 mm to 6.40 mm. An Agilent E8363B vector network analyzer in the range of 0.1–18 GHz was used to measure the complex permeability and permittivity of the coaxial samples. Moreover, the reflection coefficients with ($S_{11\text{-SHORT}}$) and without ($S_{11\text{-OPEN}}$) a backed metal plate were separately detected to study the reflection properties of the composites and calculate the electromagnetic wave energy reflected at the air-absorber interface (AA interface) and the absorber-metal plate interface (AM interface).[8,9]

According to the transmission line theory, when a wave is normally incident to a absorber layer with a backed metal plate, the RL curves at a given absorber thickness can be calculated from the complex permeability and permittivity by means of the following expressions:[10,11]

$$Z = \frac{Z_{in}}{Z_0} = \sqrt{\frac{\mu_r}{\varepsilon_r}} \tanh\left(j\frac{2\pi t}{\lambda}\right)\sqrt{\mu_r \varepsilon_r}, \quad (1)$$

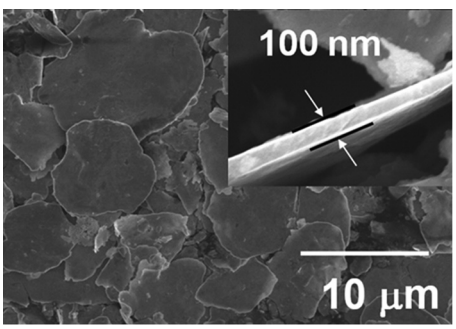


FIG. 1. The SEM image of flake-shaped carbonyl-iron particles. The inset is the profile of the flake.

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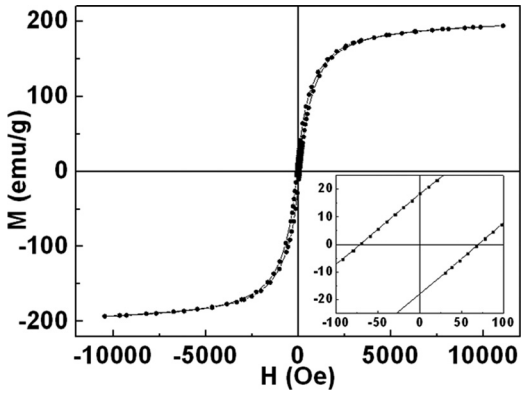


FIG. 2. The hysteresis loop of flake-shaped carbonyl-iron particles. The enlarged curve near the zero magnetic field is shown in the inset.

$$RL(dB) = 20\log\left|\frac{Z-1}{Z+1}\right|, \tag{2}$$

where Z (Z_{in}/Z_0) is the normalized input impedance related to the impedance in free space; ε_r and μ_r are the complex permittivity and permeability; λ is the wavelength in free space; and t is the thickness of the absorber.

III. RESULT AND DISCUSSION

A. Electromagnetic parameters and RL properties of FSCI particle/paraffin composite

Figure 3 shows the frequency dependence of complex permeability and permittivity for the FSCI particle/paraffin composite. It can be seen that the permittivity almost remain constant in 0.1–18 GHz range, and the magnetic spectrum exhibits a relaxation type. With the purpose of characterizing the electromagnetic wave attenuation properties, the RL curves at different absorber thickness are simulated using Eqs. (1) and (2). The results are shown in Fig. 4(a). From the RL curves, we can get the following properties: (1) no RL peak appears at a small thickness of 0.64 mm; (2) at the larger thickness, the RL peak appears and moves to lower frequency with the increase of the thickness; (3) the RL peak intensity first increases then decreases with the increase of the thickness; and (4) two RL peaks appear at a sufficiently large thickness (6.40 mm).

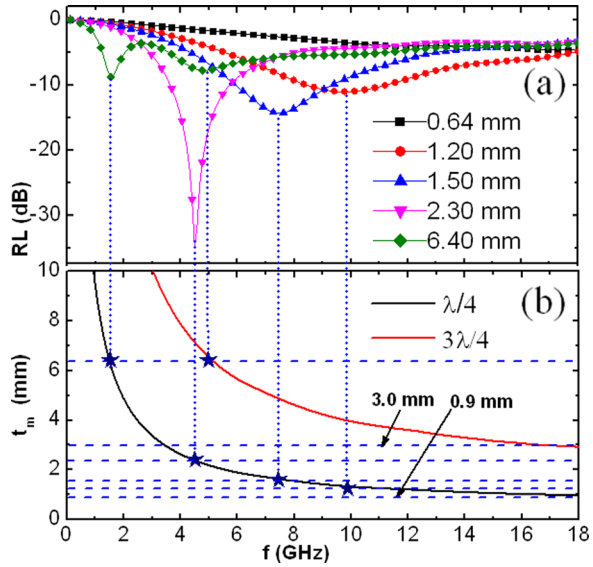


FIG. 4. Dependence of RL on frequency at various thicknesses (a) and dependence of $\lambda/4$ and $3\lambda/4$ on frequency (b) for the FSCI particle/paraffin composite from its complex permeability and permittivity with frequency. The dashed lines in (b) are thickness contours at 0.9 mm, 1.20 mm, 1.50 mm, 2.30 mm, 3.0 mm, and 6.40 mm. The vertical dot lines are extended from the RL peaks in (a), each line from the RL peak under an absorber thickness crosses with its corresponding thickness contour in (b). The crossover points are indicated by the asteroid dots.

B. Dependence of RL frequency on electromagnetic parameters and thickness

From the magnetic spectrum and dielectric spectrum of the FSCI particle composite, we can see the imaginary part of the permittivity is constant in 0.1–18 GHz, while the imaginary part of permeability has a maximum at 6 GHz. According to the electromagnetic loss model, the large ε'' and μ'' are useful to significantly absorb the electromagnetic wave energy.[12] Hence, the absorbing peak should be at 6 GHz. Even the composite thickness changes, the absorbing peak should keep at 6 GHz because the electromagnetic parameters are invariable with composite thickness. However,

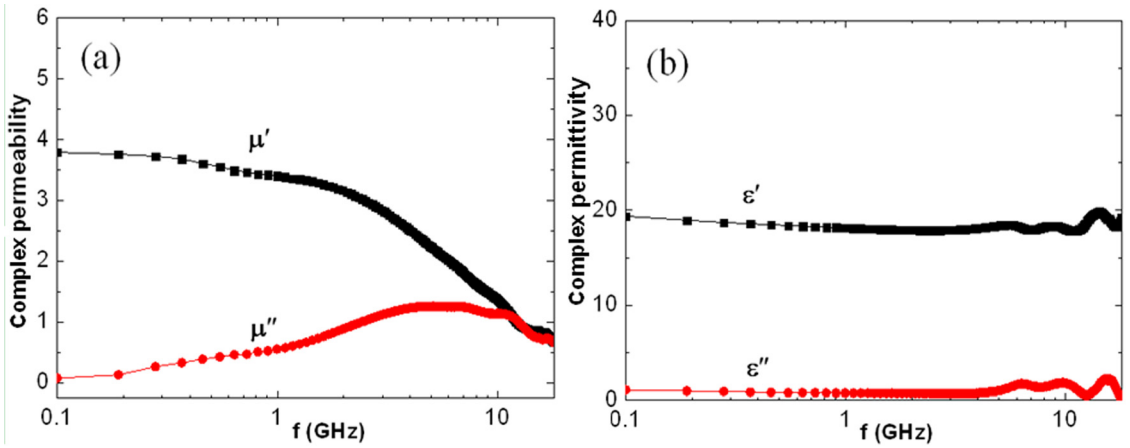


FIG. 3. Complex permeability (a) and permittivity (b) for the FSCI particle/paraffin composite with 20 vol. %.

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the simulated results show that the RL peak frequency is closely related to the thickness and moves to the lower frequency with the increase of composite thickness; moreover, the RL peak intensity first increases then decreases. All these properties cannot be explained reasonably by electromagnetic loss model.

The quarter-wavelength cancellation has been successfully used to explain the relationship between RL peak frequency and absorber thickness for Co_2Z and spherical carbonyl-iron particle composites.[9,13] The cancellation process is described as follows. When an electromagnetic wave is normally incident to an absorber backed by a perfect metal plate, it will be partially reflected at the AA interface and the other part enters the absorber. When the entered wave reaches the AM interface, it will be reflected completely and then emerges from the AA interface. The two waves cancel at interface AA interface when they are out of phase by 180°. In this case, the thickness of the absorber is satisfied with the quarter-wavelength condition

$$t_m = \frac{nc}{4f_m\sqrt{\varepsilon_r\mu_r}} \quad (n = 1, 3, 5\ldots), \tag{3}$$

where t_m and f_m are matching thickness and matching frequency of a RL peak, ε_r and μ_r are complex permittivity and permeability at f_m, and c is the velocity of light. Fig. 4(b) gives the dependence of $\lambda/4$ and $3\lambda/4$ on frequency for the FSCI particle composite. The thickness contours at 1.20 mm, 1.50 mm, 2.30 mm, and 6.40 mm, which correspond to the absorber thicknesses in Fig. 4(a) except for 0.64 mm, are plotted in this figure. In Fig. 4(a), the vertical dot lines are extended from the RL peaks, and each line from the RL peak under an absorber thickness crosses with its corresponding thickness contour in (b). The crossover points are indicated by the asteroid dots. We can clearly see that all the asteroid dots locate on the $\lambda/4$ and $3\lambda/4$ curves, which demonstrates all the absorber thicknesses are in good agreement with the quarter-wavelength of the composite when the peak appears in the RL curve. Since the RL peak comes from the quarter-wavelength cancellation, the frequency and the number of RL peak are determined by the quarter-wavelength cancellation condition when the composite thickness is fixed. Furthermore, we can get the following information in 0.1–18 GHz range from the thickness contours in Fig. 4(b): (1) when the absorber thickness $t < 0.9$ mm, the thickness contour has no crossover point with the $\lambda/4$ curve, indicating no RL peak exists in the RL curve, as shown in Fig. 4(a) at $t = 0.64$ mm; (2) when $0.9\,\text{mm} < t < 3.0\,\text{mm}$, the thickness contour has one crossover point with the $\lambda/4$ curve and one peak should exist in the RL curve; (3) when $t > 3.0$ mm, the thickness contour has crossover points with the $\lambda/4$ and $3\lambda/4$ curves, and two peaks should exist in the RL curve, which has been verified in the RL curve shown in Fig. 4(a) at 6.40 mm.

C. The mechanism for the intensity variation of RL peak

Figure 5 is the diagram of an electromagnetic wave normally incident on an absorber without (a) and with (b) a backed metal plate. For an absorber without a metal plate, the incident wave (E_0) is partially reflected at the AA interface, labeled as E_{AA} which can be detected by $S_{11\text{-OPEN}}$, partially penetrates through the layer, denoted as S_{21}, and partially attenuated in the absorber, labeled as E_{loss1}. When the absorber is backed by a metal plate, the partially reflected wave energy (E_{AA}) at the AA interface is the same with that without the metal plate. The penetrating wave in Fig. 5(a) is reflected completely at the AM interface in this situation and then emerges from the AA interface. The energy of this emerging wave is labeled as E_{AM}. The attenuated wave energy in the absorber in this case is E_{loss2}. The RL properties in Fig. 5(b) can be directly measured by $S_{11\text{-SHORT}}$, which is coincident with the calculated value from the transmission line theory. If the energy of the incident electromagnetic wave is 1, through measuring $S_{11\text{-OPEN}}$ and $S_{11\text{-SHORT}}$, we can get an important energy relationship

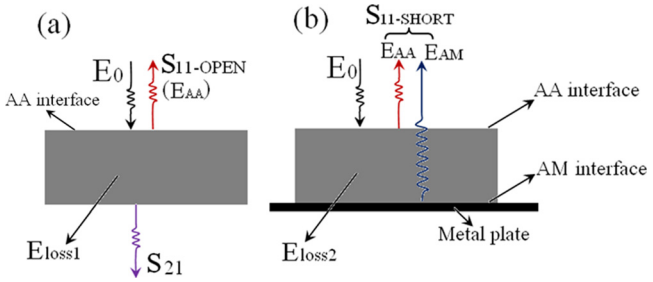


FIG. 5. The scheme of an electromagnetic wave normally incident to an absorber without (a) and with (b) a backed metal plate. E_0 is the energy of the incident wave; E_{AA} is the wave energy reflected at the AA interface without the metal plate; E_{AM} is the wave energy reflected at the AM interface with the metal plate; E_{loss1} and E_{loss2} are the attenuated wave energy inside the absorber shown in (a) and (b).

$$2E_{11-OPEN} \pm E_{11-SHORT} + E_{loss2} = 1, \tag{4}$$

where $E_{11\text{-}OPEN}$ and $E_{11\text{-}SHORT}$ are the energy ratio transformed from the measured $S_{11\text{-}OPEN}$ and $S_{11\text{-}SHORT}$ parameters. When E_{AA} is larger than E_{AM}, the plus of $E_{11\text{-OPEN}}$ and $E_{11\text{-SHORT}}$ is used; otherwise, the minus is used. From this equation, we can obtain the electromagnetic wave energy reflected at AA and AM interfaces and attenuated in the absorber.

Figure 6 shows the frequency dependence of $S_{11\text{-OPEN}}$ and $S_{11\text{-SHORT}}$ curves corresponding to the FSIC particle/paraffin composites without and with a backed metal plate under various thicknesses. The value of $S_{11\text{-OPEN}}$ for the composites is quite small in the low frequency region. It rapidly increases and remains constant at the high frequency region. The variation regularity of $S_{11\text{-OPEN}}$ with frequency reveals that only a small part of electromagnetic wave is reflected at the AA interface in the low frequency region, and the reflected wave rapidly increases and remains constant with the frequency increasing. It is noted that no RL peak is found in all the $S_{11\text{-OPEN}}$ curves at various thicknesses. This proves that the RL peak cannot appear if the composite has no backed metal plate. However, after the composite is backed by a metal plate, the RL peak is found in the $S_{11\text{-SHORT}}$ curve except for the absorber with 0.65 mm. Owing to that, the

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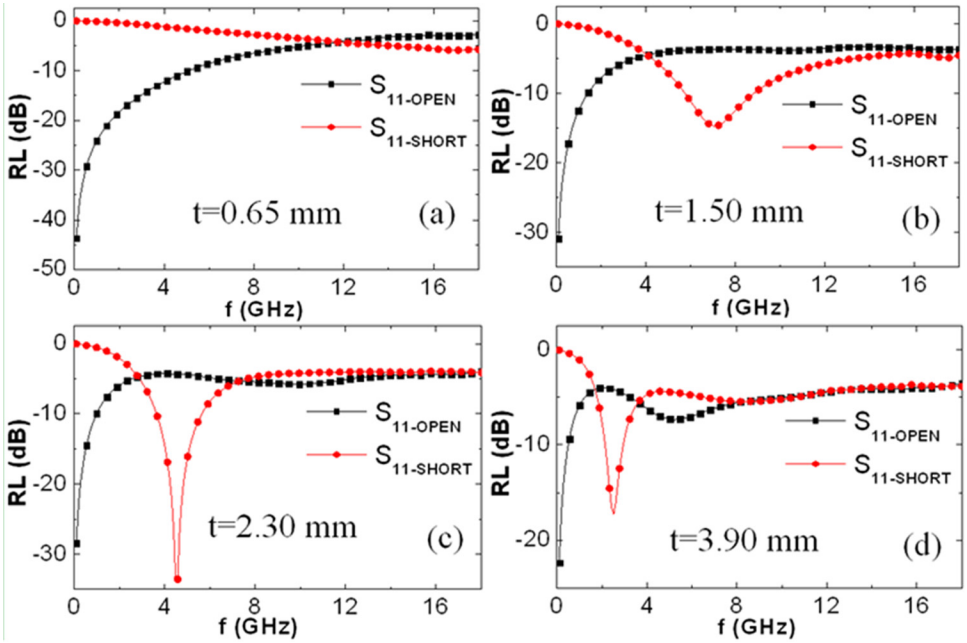


FIG. 6. The frequency dependence of $S_{11\text{-OPEN}}$ and $S_{11\text{-SHORT}}$ for FSCI particle/paraffin composite under various thicknesses: (a) 0.65 mm, (b) 1.50 mm, (c) 2.30 mm, and (d) 3.90 mm.

penetrating wave labeled as S_{21} shown in Fig. 5(a) is completely reflected at the AM interface and then emerges from the AA interface when a backed metal plate is added, it superimposes with the reflected wave from the AA interface. When the composite thickness satisfies the quarter-wavelength, the RL peak appears in $S_{11\text{-SHORT}}$ curves as a result of the cancellation of the two waves. At the thickness of 0.65 mm, the composite is too thin to satisfy the quarter-wavelength condition, so no RL peak exists in the $S_{11\text{-SHORT}}$ curve. Additionally, all the RL curves measured by $S_{11\text{-SHORT}}$ agree well with the simulated results from the complex permeability and permeability at the corresponding thickness by using Eqs. (1) and (2). This experiment visually verifies that the RL peak comes from the quarter-wavelength cancellation of two reflected waves from the AA interface and AM interface for an absorber with a backed metal plate.

Table I presents the normalized EM energy reflected by the AA interface, the AM interface, and attenuated in the absorber at the RL peak frequency, calculated from $S_{11\text{-OPEN}}$ and $S_{11\text{-SHORT}}$ parameters. At 2.30 mm, the RL has a minimum value (−35 dB) at 4.58 GHz. The calculated E_{AA} and E_{AM} are nicely equal at this frequency point. In this case, the two waves reflected from AA interface and AM interface satisfy the complete cancellation and hence resulting in a strong RL peak. The corresponding thickness and frequency at the minimum RL value are called as perfect matching thickness and frequency. For the other thicknesses, the E_{AA} and E_{AM} are not equal anymore; the two waves reflected from AA interface and AM interface cannot cancel completely, leading to the weak RL peaks. From the dependence of $S_{11\text{-OPEN}}$ and $S_{11\text{-SHORT}}$ parameters on frequency, it can be concluded that the wave energy reflected from the AA interface is smaller than that from the AM interface when frequency is smaller than the perfect matching frequency (4.58 GHz), and they are going in the opposite direction when frequency is larger than 4.58 GHz. Therefore, the intensity of the RL peak in Fig. 4(a) first increases, reaches a maximum at 4.58 GHz, and then decreases with the frequency decreasing. From the table, we can find there is always a part of electromagnetic loss (E_{loss2}) within the absorber. It increases mainly with the increase of the thickness, whereas the ratio to the incident wave energy is less and is only 26% at the perfect matching point. Additionally, the modulus of ε_r/μ_r is far from the unity at the perfect matching point. This is different from most reported viewpoints in which the equal ε_r and μ_r are necessary to obtain a good RL. However, the normalized input impedance $|Z|$ at the perfect matching point is still approximately equal to 1.

TABLE I. The electromagnetic wave energy reflected by the AA interface and the AM interface, and attenuated in the absorber at the RL peak frequency for the composites with different thicknesses; the modulus of complex permittivity and permeability ($|\varepsilon_r/\mu_r|$) and the normalized input impedance |Z| at the RL peak frequency are also listed.

t (mm)	f_m (GHz)	RL (dB)	$E_{11\text{-SHORT}}$	E_{AA}	E_{AM}	E_{loss2}	$\lvert\varepsilon_r/\mu_r\rvert$	$\lvert Z\rvert$
0.65	…	…	…	…	…	…	…	…
1.50	7.08	−16	0.034	0.433	0.399	0.168	8.25	0.70
2.30	4.58	−35	4.4E-4	0.370	0.370	0.260	6.87	0.95
3.90	2.51	−17	0.019	0.347	0.366	0.287	5.66	1.49

D. The mechanism for the bandwidth of RL peak

As the RL peak comes from the cancellation of two waves, their phases should be out of π at the peak frequency point. When the frequency deviates from the peak frequency point by Δf, the peak intensity will decrease due to that the phase difference of the two reflected waves deviates from π to π-$\Delta\theta$. Figure 7 shows the bandwidth scheme of a RL peak. E_P is the energy at the peak frequency point, and E_d is the energy at the designated frequency point. As E_P is the subtraction of E_{AA} and E_{AM} with the phase difference of π, it can be described by

$$E_p = E_{AA}\cos^2 0 - E_{AM}\cos^2\pi = E_{AA} - E_{AM}. \quad (5)$$

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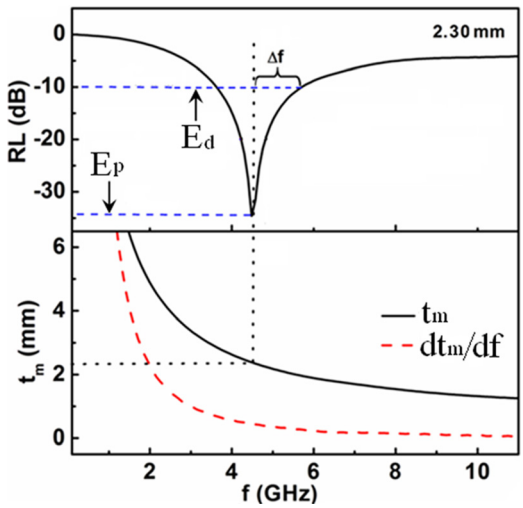


FIG. 7. The frequency dependent of quarter-wavelength and the scheme of bandwidth of RL peak for FSCI particle/paraffin composite. dt_m/df is the derivation of matching thickness to frequency; E_P is the energy at the peak frequency point, and E_d is the energy at -10 dB; Δf is the half bandwidth at -10 dB.

E_d is the subtraction of E_{AA} and E_{AM} with the phase difference of $\pi - \Delta\theta$ and can be written as

$$E_d = E_{AA}\cos^2 0 - E_{AM}\cos^2(\pi - \Delta\theta). \tag{6}$$

Hence, the energy difference of E_P and E_d is

$$E_d - E_P = E_{AM}(1 + \cos^2\Delta\theta). \tag{7}$$

Assuming that $\Delta\theta$ is small enough, it can be described by

$$\cos\Delta\theta = 1 - \frac{(\Delta\theta)^2}{2!} \approx 1 - \frac{1}{2}(\Delta\theta)^2, \tag{8}$$

so we can get

$$\Delta\theta = \left[2\left(1 - \left(1 - \frac{E_d - E_p}{E_{AM}}\right)^{1/2}\right)\right]^{1/2}. \tag{9}$$

According to the origin of bandwidth, $\Delta\theta$ can also be written as

$$\Delta\theta = \frac{\pi}{t_m}\left[\left(\left|\frac{dt_m}{df}\right|\right)\cdot\Delta f\right]. \tag{10}$$

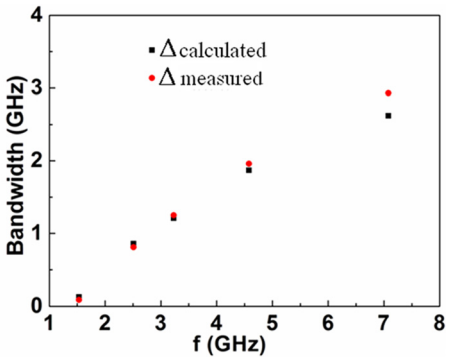


FIG. 8. The measured and calculated bandwidth at $E_d = -10$ dB for FSCI particle/paraffin composite under various frequencies.

TABLE II. The parameters of the measured and calculated bandwidth at $E_d = -10$ dB for FSCI particle/paraffin composite under various thicknesses.

t (mm)	f_m (GHz)	dt_m/df	$\Delta_{calculated}$ (GHz)	$\Delta_{measured}$ (GHz)
1.50	7.08	0.17	2.62	2.93
2.30	4.58	0.43	1.87	1.96
3.01	3.23	0.86	1.21	1.25
3.90	2.51	1.42	0.86	0.81
6.40	1.53	3.92	0.13	0.09

Therefore, the bandwidth can be expressed by

$$\Delta = 2\Delta f = \left[2\left(1 - \left(1 - \frac{E_d - E_p}{E_{AM}}\right)^{1/2}\right)\right]^{1/2}\cdot\frac{2t_m}{\pi}\Big/\left|\frac{dt_m}{df}\right|. \tag{11}$$

Equation (11) demonstrates that the bandwidth of a RL peak is influenced by the derivation of matching thickness to frequency, the matching thickness, the reflected wave energy from AM interface, the energy difference of peak frequency, and designated frequency. In all the factors, the derivation of matching thickness to frequency seems most sensitive to the bandwidth.

Figure 8 shows the measured and calculated bandwidth at $E_d = -10$ dB under various frequencies. The parameters of the bandwidth are shown in Table II. Obviously, the bandwidths from the theoretical calculations under various thicknesses are almost in accordance with those of the experimental results. This result demonstrates the validity of the derived bandwidth formula from the quarter-wavelength cancellation mechanism.

IV. CONCLUSION

In this work, we deeply investigated the reflection loss properties for the flake-shaped carbonyl-iron particle/paraffin composite. The relationship between the RL peak frequency and the absorber thickness was successfully explained by the quarter-wavelength cancellation. This absorbing mechanism was visually verified by measuring the reflection coefficient without and with a backed metal plate. Through analyzing and calculating the energy of the electromagnetic waves reflected from the air-absorber and absorber-metal plate interfaces, it was concluded that the intensity of the reflection loss peak is determined by the energy difference of the two waves. When the energy of the two reflected waves is equal, the intensity of the reflection loss peak is strongest, otherwise it becomes weak. The bandwidth formula of the reflection loss peak was derived from the quarter-wavelength cancellation mechanism and the bandwidth is sensitively related to the derivation of matching thickness to frequency.

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